



Knowledge Graphs

Lecture 5 - Knowledge Graph Applications

5.3 RDF and OWL Knowledge Graphs

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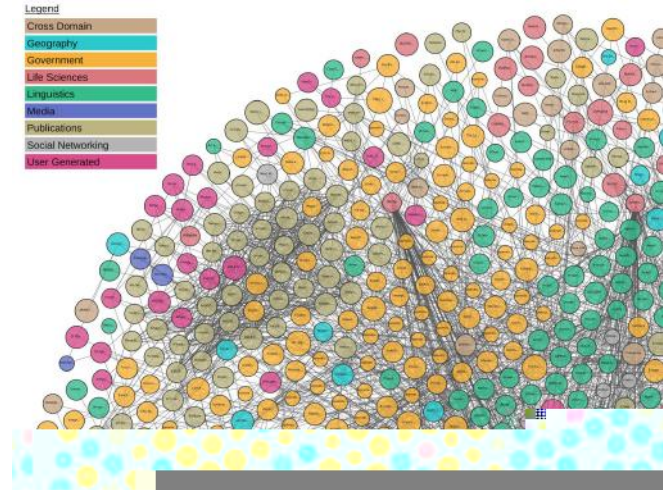
Knowledge Graphs

Lecture 5: Knowledge Graph Applications

5.3 RDF and OWL Knowledge Graphs

Creating Knowledge Graphs

- Curated approaches:
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- Collaborative approaches:
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- Automated semi-structured approaches:
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- Automated unstructured approaches:
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- Linking existing datasets:
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Creating Knowledge Graphs

- Curated approaches:

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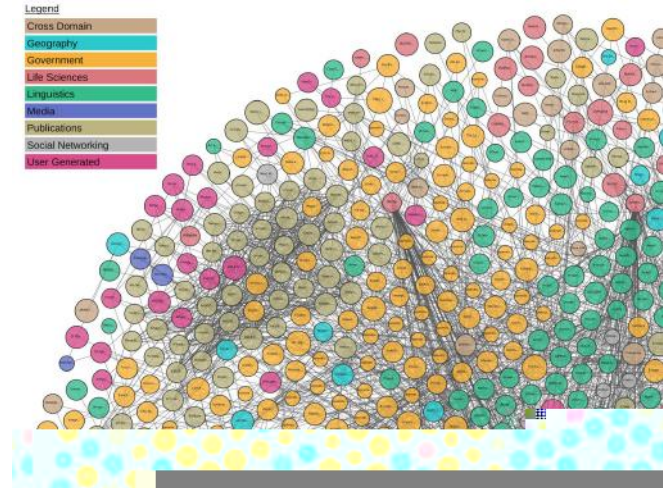
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Creating Knowledge Graphs

- Collaborative approaches:

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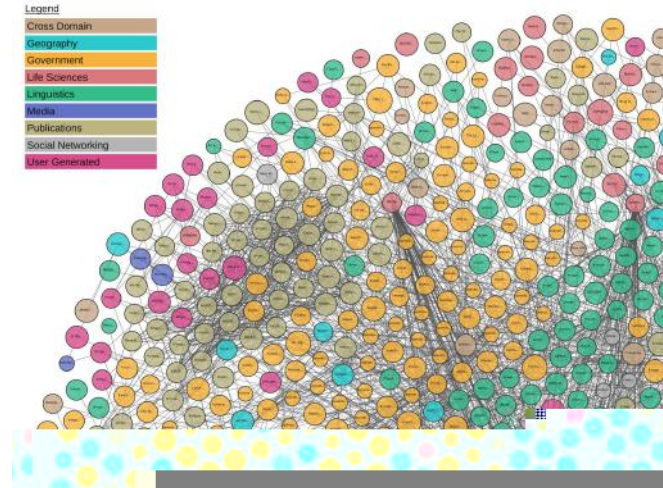
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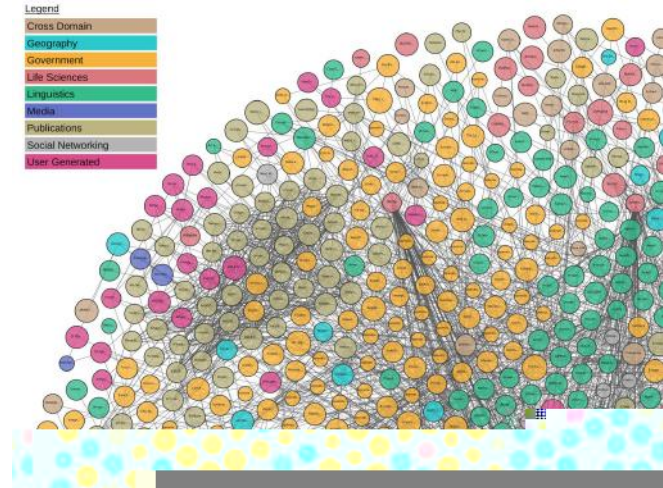
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Creating Knowledge Graphs

- Automated semi-structured approaches:

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Creating Knowledge Graphs

- Automated unstructured approaches:

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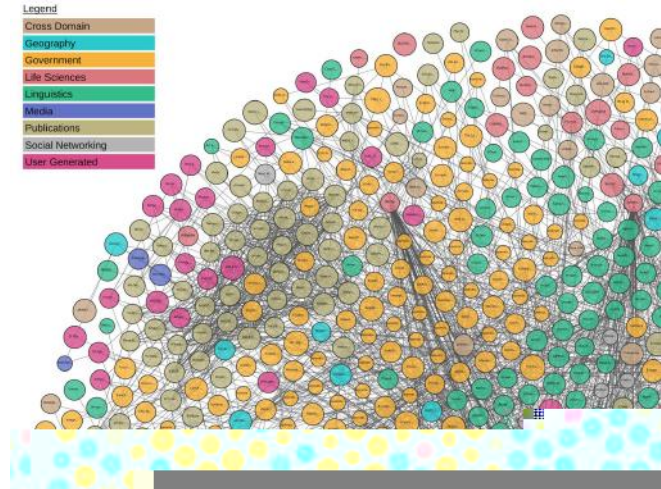
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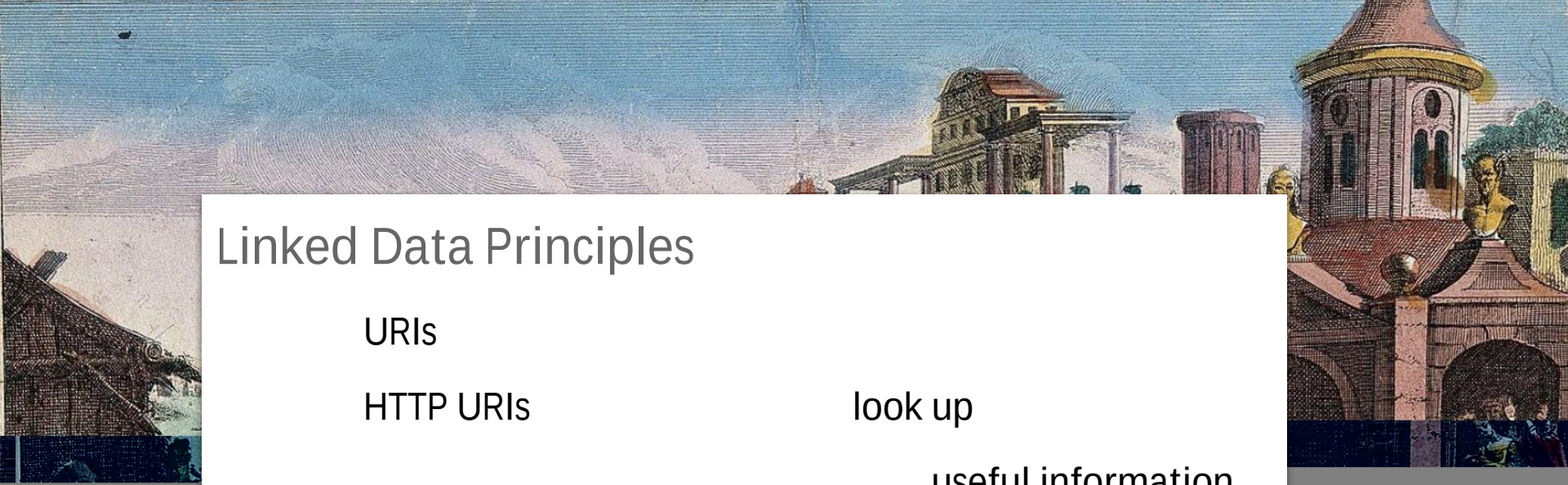
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- Noise



<http://lod-cloud.net/>



Linked Data Principles

URIs

HTTP URIs

look up

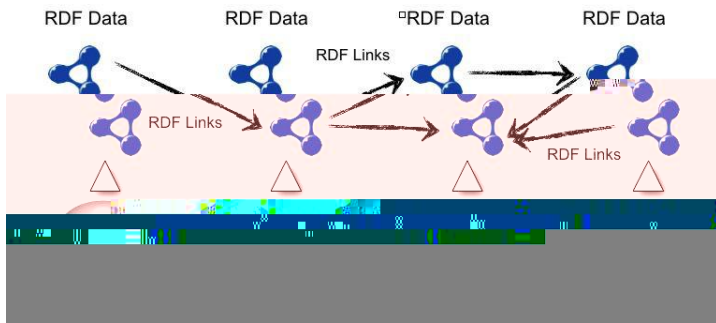
useful information

standards

links to other URIs

Advantages of Linked Open Data vs. APIs

- Simple generic API
simple reuse data sharing
- RDF Data model extensibility
- Transport via http
- Ontologies meaningful connections
- Reasoning generate new knowledge



Popular (Open) Knowledge Graphs

	DBpedia	YAGO	Wikidata	OpenCyc	NELL
Version	2016-04	YAGO3	2016-08-01	2016-09-05	08m.995
# instances	5,109,890	5,130,031	17,581,152	118,125	1,974,297
# axioms	397,831,457	1,435,808,056	1,633,309,138	2,413,894	3,402,971
avg. indegree	13.52	17.44	9.83	10.03	5.33
avg. outdegree	47.55	101.86	41.25	9.23	1.25
# classes	754	576,331	30,765	116,822	290
# relations	3,555	93,659	11,053	165	1,334
Releases	biyearly	> 1 year	live	> 1 year	1-2 days

Towards Profiling Knowledge Graphs



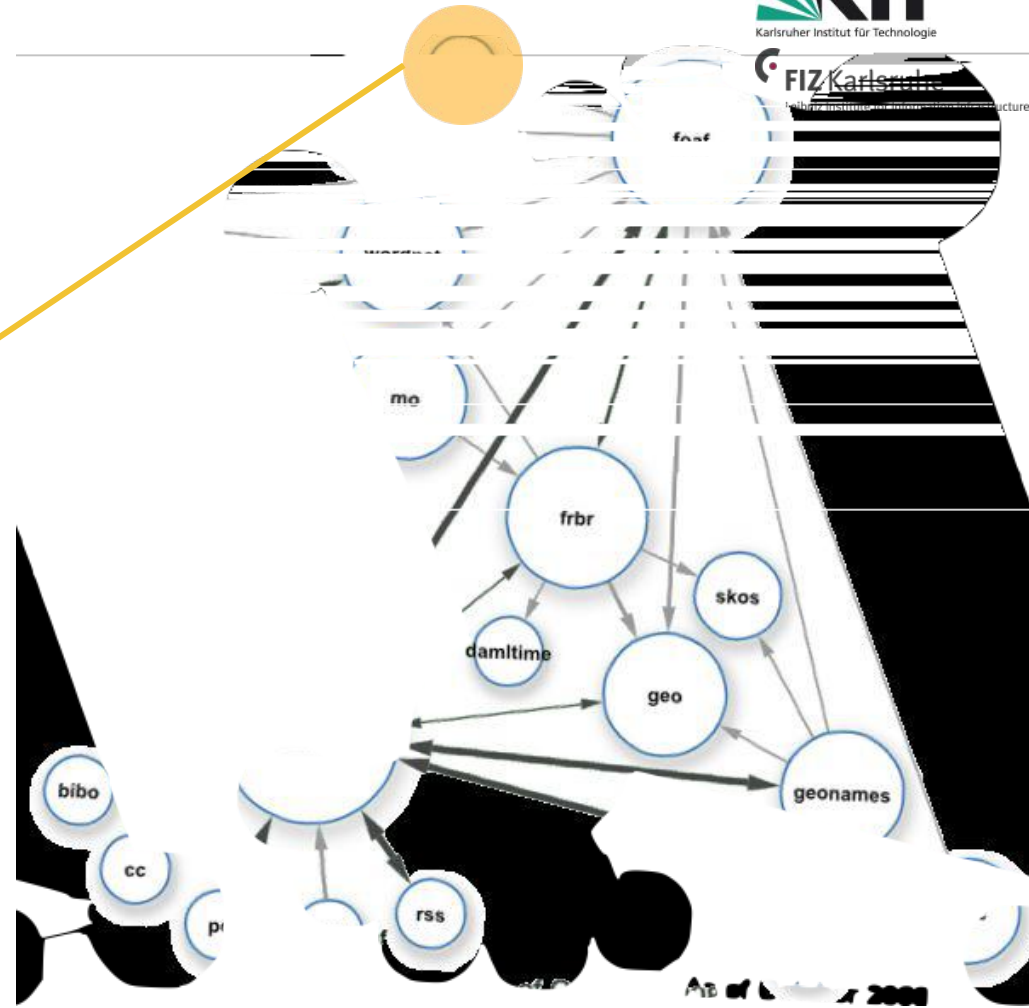
Popular (Proprietary) Knowledge Graphs

	Data model	Size of the graph	Development stage
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
Facebook	All of the attributes and relations are structured and strongly typed, and optionally indexed to enable efficient retrieval, search, and traversal.	~50 million primary entities, ~500 million assertions	Actively used in products
eBay	Entities and relation, well-structured and strongly typed	Expect around 100 million products, >1 billion triples	Early stages of development and deployment
IBM	Entities and relations with evidence information associated with them.	Various sizes. Proven on scales documents >100 million, relationships >5 billion, entities >100 million	Actively used in products and by clients

Linked Data Ontologies

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- OWL
owl:sameAs

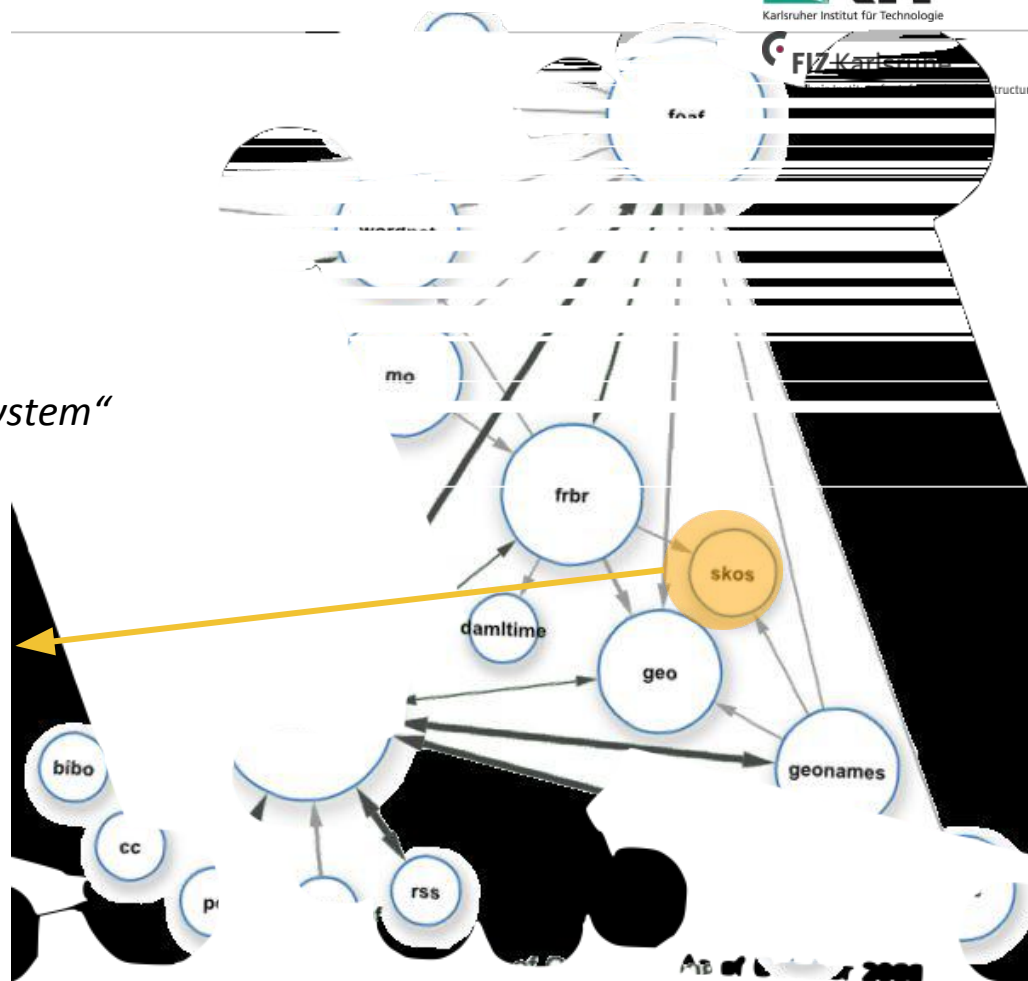
owl:equivalentClass



Linked Data Ontologies

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- **SKOS**
 - „Simple Knowledge Organization System“
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- `skos:Concept`
- `skos:narrower`
- `skos:broader`
- `skos:related`
- `skos:exactMatch`,
`skos:narrowMatch`,
`skos:broadMatch`,
`skos:relatedMatch`



Linked Data Ontologies

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- **UMBEL**

- „Upper Mapping and Binding Exchange Layer“

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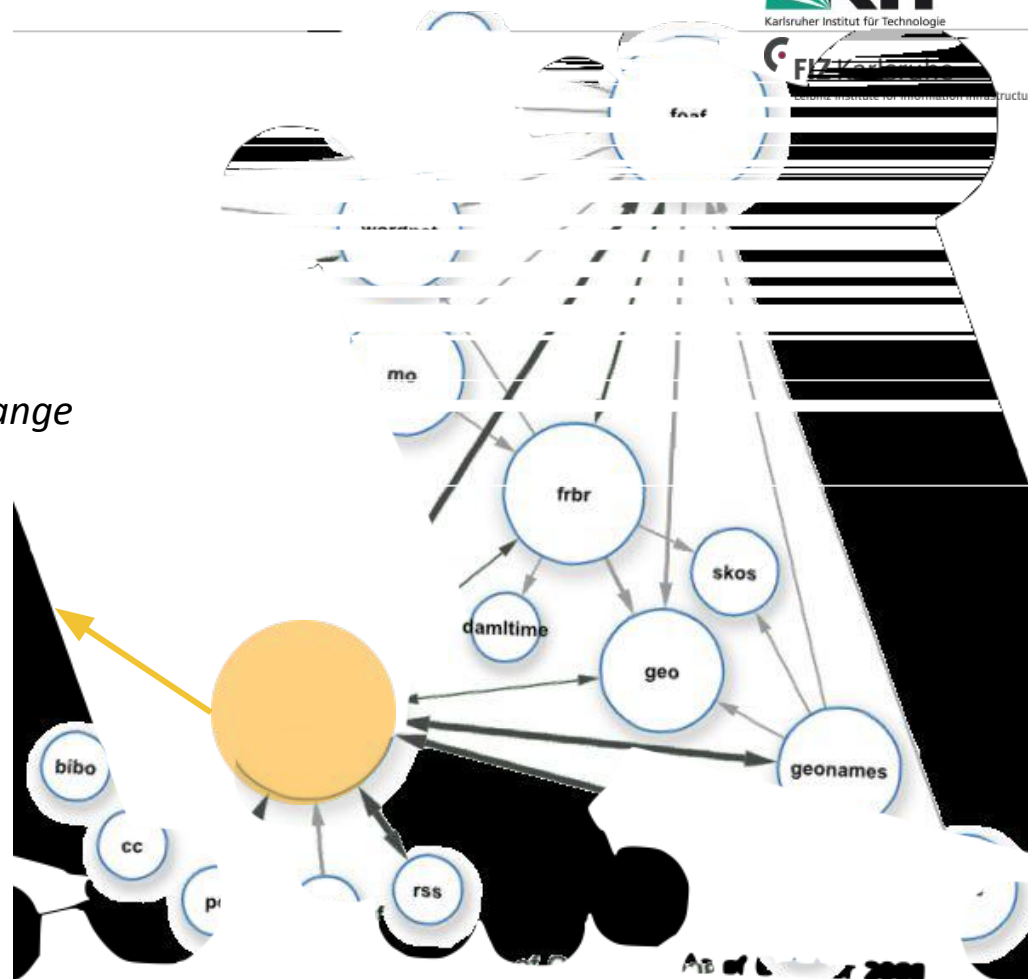
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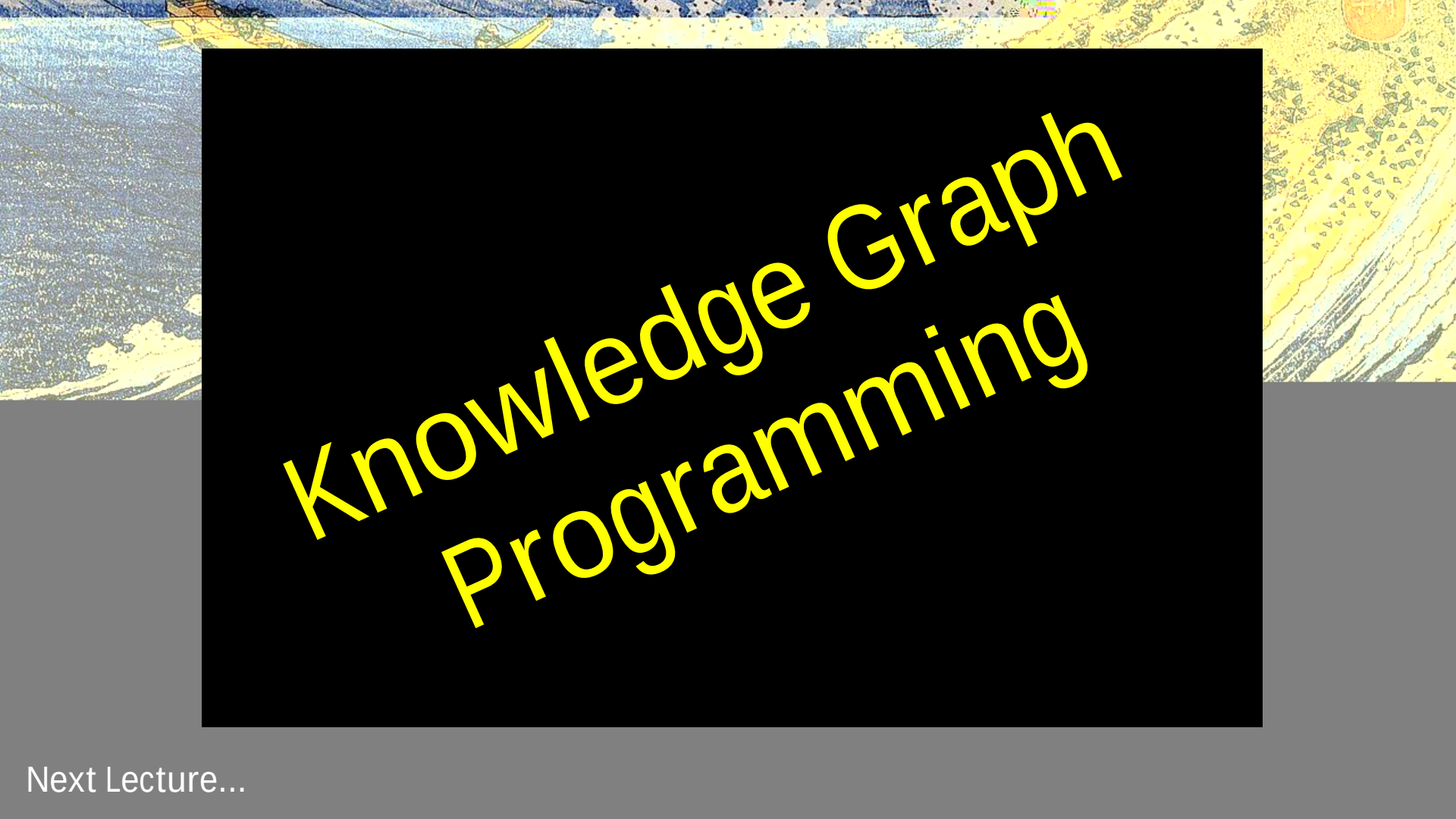
- skos:Concept

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- owl:equivalentClass
rdfs:subClassOf

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Knowledge Graph Programming

Next Lecture...

Picture References:

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